

Table S2. Regional model performance across all model variants. RMSE values in % and R^2 for each region. Overall performance shown in highlighted rows, followed by regional breakdowns. Models with '.sp' suffix use spatial GP.

Model	Region	N	RMSE	R^2
baseline	Overall	818	21.1	0.703
	Africa	133	20.2	0.013
	Americas	221	14.9	0.902
	Asia	391	22.0	0.343
	Europe	32	15.6	0.700
	Oceania	41	39.8	0.676
baseline_sp	Overall	818	15.5	0.840
	Africa	133	11.7	0.594
	Americas	221	10.0	0.954
	Asia	391	19.1	0.467
	Europe	32	12.2	0.753
	Oceania	41	13.6	0.816
baseline_env	Overall	818	19.6	0.743
	Africa	133	19.2	0.057
	Americas	221	15.2	0.896
	Asia	391	20.9	0.373
	Europe	32	13.6	0.730
	Oceania	41	30.3	0.691
baseline_env_sp	Overall	818	14.8	0.854
	Africa	133	11.3	0.627
	Americas	221	10.2	0.952
	Asia	391	18.1	0.524
	Europe	32	12.0	0.763
	Oceania	41	13.4	0.813
baseline_veg	Overall	818	20.4	0.721
	Africa	133	16.2	0.251
	Americas	221	17.1	0.873
	Asia	391	21.5	0.374
	Europe	32	17.2	0.662
	Oceania	41	35.5	0.679
baseline_veg_sp	Overall	818	15.1	0.848
	Africa	133	10.7	0.663
	Americas	221	10.0	0.954
	Asia	391	18.8	0.489
	Europe	32	12.7	0.730
	Oceania	41	13.0	0.821
full	Overall	818	19.3	0.751
	Africa	133	16.1	0.244
	Americas	221	16.8	0.873
	Asia	391	20.6	0.392
	Europe	32	14.2	0.709
	Oceania	41	29.4	0.648
full_sp	Overall	818	14.6	0.858
	Africa	133	10.3	0.689
	Americas	221	10.0	0.953
	Asia	391	17.9	0.533
	Europe	32	12.5	0.740
	Oceania	41	13.0	0.820
full_interact	Overall	818	19.3	0.751
	Africa	133	16.2	0.242
	Americas	221	16.8	0.873
	Asia	391	20.6	0.391
	Europe	32	14.3	0.708
	Oceania	41	29.4	0.645
full_interact_sp	Overall	818	14.6	0.858
	Africa	133	10.3	0.689
	Americas	221	10.1	0.953
	Asia	391	17.9	0.533
	Europe	32	12.6	0.738
	Oceania	41	13.0	0.819
elevation_only_sp	Overall	818	14.8	0.853
	Africa	133	11.3	0.627
	Americas	221	10.2	0.952
	Asia	391	18.1	0.523
	Europe	32	12.0	0.762
	Oceania	41	13.4	0.814
elevation_c4_sp	Overall	818	14.8	0.854
	Africa	133	11.3	0.625
	Americas	221	10.1	0.952
	Asia	391	18.1	0.524
	Europe	32	12.0	0.761
	Oceania	41	13.2	0.816
c4_only_sp	Overall	818	15.5	0.839
	Africa	133	12.0	0.574
	Americas	221	10.1	0.953
	Asia	391	19.2	0.466
	Europe	32	12.3	0.750
	Oceania	41	13.4	0.822
elevation_c4_interact_sp	Overall	818	14.8	0.853
	Africa	133	11.3	0.622
	Americas	221	10.1	0.952
	Asia	391	18.1	0.523
	Europe	32	12.0	0.761
	Oceania	41	13.3	0.816